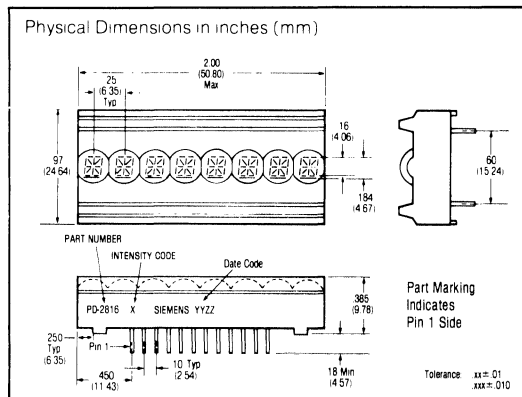
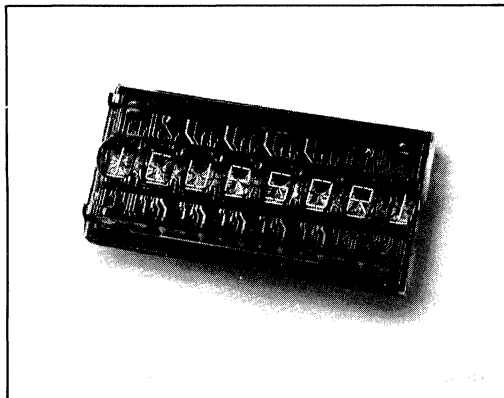


.160" RED, 8 DIGIT, 18 SEGMENT INCLUDING DECIMAL ALPHANUMERIC Programmable Display™ With Built In CMOS Control Functions

Preliminary



FEATURES

- Microprocessor Compatible
- End Stackable, 8-Character Package
160 Mil High, Magnified Monolithic Char.
- Viewing Angle $\pm 32^\circ$
- 64 Character ASCII Format
- 18-Segment Including Underline and Decimal
- Control & Display Memory Read/Write
- Total Read/Write Time: 200 ns min.
- Built-in Character Generator
- Built-in Multiplex and LED Drive Circuitry
- Software Controlled Features:
 - Programmable Highlight Attribute (Blinking, Non-blinking, Underline)
 - Asynchronous Memory Clear Function
 - Lamp Test
 - Display Blank Function
 - Single or Multiple Character Blinking Function
 - Character Underline Function
 - Programmable Intensity, 3 Brightness Levels
- Intensity Coded For Display Uniformity
- TTL Compatible, Single 5 Volt Power
- Asynchronous Access to Each Digit
- Easily Cascaded
- Internal Or External Clock Source
- Lower CPU Overhead
- Rugged Encapsulated Package

GENERAL DESCRIPTION

The PD-2816 is an 8 digit 18 segment Programmable Display. It is designed to ease the job of interfacing to microcomputer systems by allowing all internal memory and registers to be addressable through the 8 bit bi-directional data bus. A control register accessible through the data bus controls all features of the display. Some of these features include the Highlight Attribute (Blinking, Nonblinking, Underline) and programmable intensity.

The heart of the display device is a built-in CMOS integrated circuit. This integrated circuit contains memory, ASCII ROM character generator, multiplexing circuitry, display drivers, and bus control circuitry. Each display digit is directly addressable and includes a Highlight Attribute control bit. A display system can be built using any number of PD-2816's cascaded together.

The display itself consists of eight 18 segment, 0.160" high characters. Each character contains a decimal point and an underline segment. All displays are intensity coded for ease of brightness matching in multiple module designs.

For further information, refer to PD-2816 data sheet supplement.

Important: Refer to Appnote 18, "Using and Handling Intelligent Displays". Since this is a CMOS device, normal precautions should be taken to avoid static damage.

Specifications are subject to change without notice.

OPTOELECTRONIC CHARACTERISTICS AT 25°C

MAXIMUM RATINGS

Input Voltage Relative to Gnd
(all inputs) -0.5 to $V_{CC} + 0.5$ Vdc
Operating Temperature -20°C to 70°C
Storage Temperature -20°C to 70°C

OPTICAL CHARACTERISTICS

Spectral Peak Wavelength 655nm Typ
Spectral Line Half Width 40nm Typ
Viewing Angle $+/- 32^{\circ}$
Digit Height 160 mils
Luminous Intensity @ $V_{CC} = 5\text{V}$ 0.15 mcd/Seg
(@ 100% Intensity)
Intensity matching
Seg to Seg @ $V_{CC} = 5\text{V}$ 1.8:1

D.C. CHARACTERISTICS

Parameters	Conditions	Min.	Typ.	Max.	Units
V_{CC}		4.5		5.5	Volts
I_{CC} (Display Blank)	$V_{CC} = 5\text{V}$ $WR = V_{CC}$ $V_{IN} = 0\text{V}$	2.0	5.0	10	mA
I_{CC} (10 segs /char 8 digits on)	@ $V_{CC} = 5\text{V}$	80	125	150	mA
VIL (All inputs)	@ $V_{CC} = 5\text{V}$	-0.5		0.8	Volts
VIH (All inputs) 1)	@ $V_{CC} = 5\text{V}$	3.0			Volts
IIL (All inputs)	@ $V_{CC} = 5\text{V}$ $V_{IN} = 8\text{V}$			400	μA
CLK Drive CLK I/O Output 2)	@ C_{IN} T5pF / Input	1		6	Devices (PD-2816)

1) VIH Min = 60% V_{CC}

2) See "CASCAIDING" for explanation

SWITCHING SPECIFICATIONS (@ 25°C AND $V_{CC} = 4.5\text{V}$)

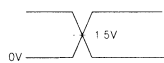
READ CYCLE TIMING

Parameter	Description	Specification (ns)
TAS	Address Set up	25 min
TRD	Read pulse	100 min
TDD	Delay before data is valid	150 max
TDH	Data hold valid after RD	50 typ
TRC	Total cycle time	200 min

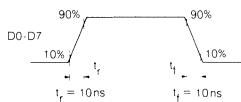
WRITE CYCLE TIMING

Parameter	Description	Specification (ns)
TWD	Write delay after CE1, CE2, and Data Stable	50 min
TWR	Write pulse	100 min
TDH	Data hold valid after WR	50 min
TWC	Total write cycle	200 min

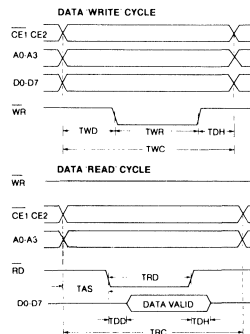
TIMING MEASUREMENT LEVELS



DATA BUS OUTPUT TRANSITIONS AT 25°C CL=150pF.



TIMING AT 25°C



NOTES

Note 1 Off Axis Viewing Angle is here defined as the minimum angle in any direction from the normal to the display surface at which any part of any segment in the display is not visible

Note 2 The display contains a CMOS integrated circuit. Normal CMOS handling precautions should be taken to avoid damage due to high static voltages or electric fields

Note 3 Unused inputs must be tied to an appropriate logic voltage level (either V_{CC} or GND)

Note 4 **Warning** — Do not use solvents containing alcohol

CONTROL WORD REGISTER

The control word is addressed by holding line A3 low. The states of the other 3 address lines (A0-A2) do not matter. The control word can be read from or written to. The truth table defines each of the bits and their functions.

Bits D0 and D1 control the display brightness. Bits D2, D3 and D4 control the Highlight Attribute function. Bit D5 controls blinking. Bit D6 is a lamp test bit. Bit D7 clears the memory display.

HIGHLIGHT ATTRIBUTE FUNCTION

In the control word bits D2, D3, and D4 control the Highlight Attribute (Blinking, Non-blinking, Underline).

To control this function, a high must be present on D4.

DISPLAY BRIGHTNESS

The display can be programmed to vary between 25%, 50%, and full brightness. Bits D0 and D1 control the brightness.

DISPLAY BLINKING

The designer has the option of displaying several message priorities by blinking either the character or the underline or both. The entire display can be blinked by writing a high into bit D5 of the control word. This function is independent of the bits D2, D3, & D4. Any character can be blinked by loading the underline and using the proper Highlight Attribute code. Display blinking is approximately at 2Hz.

LAMP TEST

In the control word bit D6 is the Lamp Test bit. In order to limit peak power this sets all segments to a 50% brightness level regardless of what is in the display memory. Setting this bit has no effect on the display memory and clearing it will restore the display to its original condition.

GENERAL FUNCTION

A3	D7	Function
L	L	Various display functions — control word
L	H	Clear
H	L	ASCII character set
H	H	ASCII character set plus underline

Note: Table also applies to control word register

CONTROL WORD TRUTH TABLE

D7	D6	D5	D4	D3	D2	D1	D0	OPERATION
L	L	X	X	X	X	L	L	BLANK
L	L	X	X	X	X	L	H	25% Brightness
L	L	X	X	X	X	H	L	50% Brightness
L	L	X	X	X	X	H	H	FULL Brightness
L	L	L	H	L	L	B	B	SOLID CHARACTER SOLID UNDERLINE
L	L	L	H	L	H	B	B	BLINK CHARACTER SOLID UNDERLINE
L	L	L	H	H	L	B	B	SOLID CHARACTER BLINK UNDERLINE
L	L	L	H	H	H	B	B	BLINK CHARACTER BLINK UNDERLINE
L	L	L	L	X	X	B	B	BLINK UNDERLINE DISABLE HIGHLIGHT ATTRIBUTE
L	L	H	X	X	X	B	B	BLINKING DISPLAY (8 Digits)
L	H	L	X	X	X	X	X	LAMP TEST (50% Brightness)
H	L	L	L	L	L	L	L	CLEAR

X = DON'T CARE B = Depending on the selected brightness

DISPLAY BLANKING

The display can be blanked in one of two ways. The first is to clear all display memory locations by writing a high to bit D7 of the control word. This will "clean the slate" and prepare for new data to be displayed. The data in the RAM is cleared. The bit is automatically cleared after the display is cleared.

The second method is a non-destructive method where the display is blanked temporarily and restored again. This is accomplished by writing low in bits D0 and D1 of the control register. The display will be blank until a high is written to either or both bits.

DATA INPUT

The eight words of memory corresponding to the eight display digits are addressed through the address lines (A0-A3) and the chip enable lines (CE1 and CE2). Address bits A0-A2 address the digits 0 (right most digit) to digit 7 (left most digit). Address bit A3 is held high to address display memory, a low on A3 accesses the control word. Display data is in the 7-bit ASCII format (bits D0-D6). The character set chart shows the resulting font. With the Highlight Attributes (bits D2, D3, & D4) a combination of nonblinking, blinking and underline can be controlled independent of the digit position.

The underline (cursor) is written into the display memory by adding bit D7 to the 6 bit ASCII code of the character. To display the underline, one of the Highlight Attribute control words has to be used see Control Word Truth Table.

READ/WRITE CONTROL ADDRESS TABLE

SIGNALS									OPERATION
CE1	CE2	RD	WR	A3	A2	A1	A0		
L	H	H	H	X	X	X	X		NO OPERATION
X	X	L	L	X	X	X	X		ILLEGAL
L	H	L	H	H	L	L	L		DIGIT 0 (RIGHT)
L	H	L	H	H	.	.	.		DIGIT 1
L	H	L	H	H	.	.	.		DIGIT 2
L	H	L	H	H	H	H	H		DIGIT 3 (LEFT)
L	H	L	H	L	X	X	X		READ CONTROL REGISTER
L	H	L	H	L	L	L	L		DIGIT 0 (RIGHT)
L	H	L	H	L	.	.	.		DIGIT 1
L	H	L	H	L	H	H	H		DIGIT 3 (LEFT)
L	H	L	H	L	X	X	X		WRITE CONTROL REGISTER
L	H	L	L	X	X	X	X		DIGIT 0 (RIGHT)

X = DON'T CARE

FUNCTIONAL DESCRIPTION

The PD-2816 block diagram includes the major logic blocks and internal registers. Display memory consists of a 9×8 bit RAM block. Each of the eight 8-bit words holds the 7-bit ASCII data (bits D0-D6) and 1-bit (bit D7) for underlining each character. The ninth 8-bit memory word is used as control register. A detailed description of the control register and its functions can be found under the heading Control Word Register. Each 8-bit word is addressable and can be read from or written to.

There are five major blocks in addition to the memory. The first is the control word decoder

and control logic which dictates all of the special features of the display device. These are discussed under the various headings on the previous page.

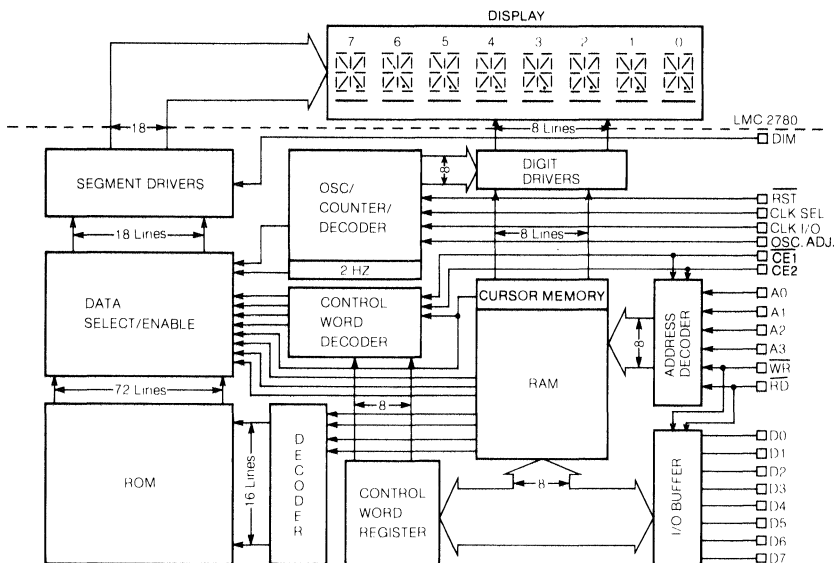
The second block is the character generator ROM. This ROM converts the 7-bit ASCII data into the proper segment configuration for the 64 characters as shown in the character set chart.

The third block is the display multiplexer and timing logic. The clock source can be either from the internal clock or from an external source (usually from the output of another PD-2816 in a multiple module display). The multiplexer controls all display output to the digit drivers so no additional logic is required for a display system.

The fourth block is the display drivers themselves. The segment and digit drivers are located on the CMOS IC and connected directly to the LEDs.

The fifth block is the LEDs. Each of the eight digits is comprised of 16 segments which make up the alpha-numeric characters, one decimal point, and an underline segment. The intensity of the display can be varied by the control word to Blank, 25%, 50%, and full brightness.

PD 2816 BLOCK DIAGRAM



MICROPROCESSOR INTERFACE

The interface to the microprocessor is through the address lines (A0-A3), the data bus (D0-D7), two chip select lines ($\overline{\text{CE1}}$, CE2), and the read ($\overline{\text{RD}}$) and write ($\overline{\text{WR}}$) lines.

Two chip enable lines are provided to simplify address decoding. $\overline{\text{CE1}}$ must be low, while CE2 must be high for any read or write operation to take place.

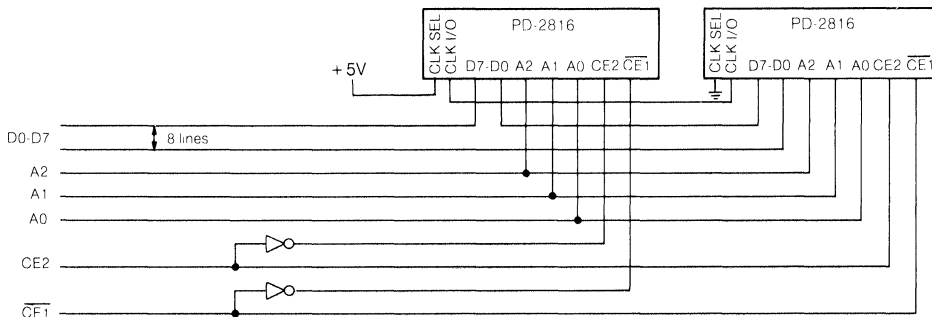
The read and write lines are both active low. During a valid read (ie: chip enable and read low) the data input lines (D0-D7) become output. A valid write will enable the data as input lines.

The address lines determine which RAM or register position will be read or written. If A3 is high then A0-A2 determine the display RAM position. If A3 is low then the operation will be to the control register regardless of the A0-A2 address lines.

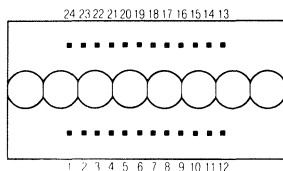
CASCADING

Cascading PD2816's is a simple operation. The requirements for cascading are: 1) decoding the correct address to determine the chip select for each additional device. 2) Selecting one display as the clock source and setting all others to accept clock input (The reason for cascading the clock is to synchronize the flashing of multiple displays). One display as a source is capable of driving 6 other PD2816's (with each input having 15pf input capacitance). If more displays are required a buffer will be necessary.

CASCADING TWO PD-2816's



TOP VIEW



PIN ASSIGNMENTS

Pin	Function	Pin	Function
1	$\overline{\text{RST}}$ RESET	13	DIM DIMMER
2	A0 ADDRESS LSB	14	WR WRITE
3	A1 ADDRESS	15	D0 DATA I/O LSB
4	A2 ADDRESS MSB	16	D1 DATA I/O
5	A3 MODE SELECT	17	D2 DATA I/O
	D0-D7	18	D3 DATA I/O
6	$\overline{\text{CE1}}$ CHIP SELECT	19	D4 DATA I/O
7	CE2 CHIP SELECT	20	D5 DATA I/O
8	CLK CLOCK I/O	21	D6 DATA I/O
9	CKS CLOCK SELECT	22	D7 DATA I/O MSB
10	$\overline{\text{RD}}$ READ	23	VCC
11	OA OSC ADJUST	24	VCC
12	GND		

PIN DEFINITIONS

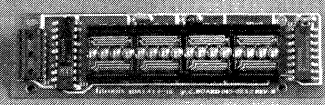
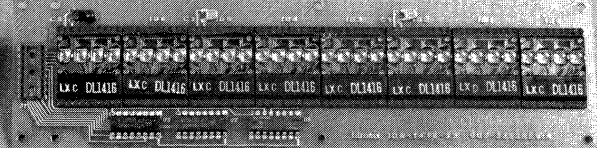
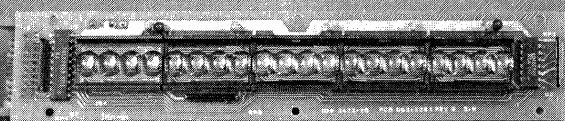
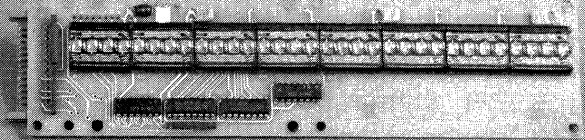
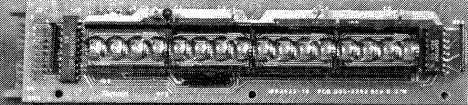
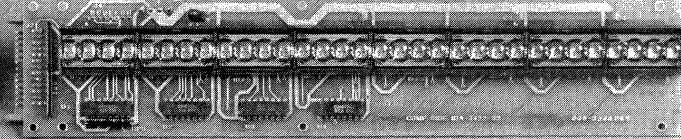
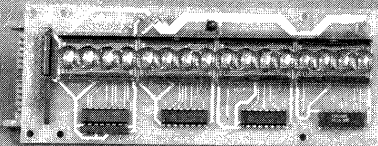
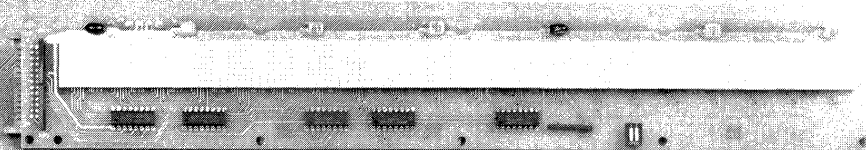
Pin	Function	Description	Pin	Function	Description
1	$\overline{\text{RST}}$	Active low reset input. Initializes multiplex counter.	12	GND	Ground
2-4	A0 A2	Address inputs for display memory RAM	13	DIM	Hardware display brightness control. When connected through external resistor to VCC, can dim display brightness. For normal operation leave open.
5	A3	Selects whether read/write from/to display memory (high) or the control register (low).	14	$\overline{\text{WR}}$	Active low write enable input. If the display is selected a low will write the data on the data bus into the selected register or memory.
6	$\overline{\text{CE1}}$	Active low chip enable input	15-22	D0-D7	Data Bus. The data bus lines are bidirectional tri state signals connected to the system data bus. The outputs are enabled during a read operation of the display memory or the control register. The outputs are disabled and the inputs read during a write cycle to the display memory or the control register.
7	CE2	Active high chip enable input	23-24	VCC	+5 volt supply — both must be connected.
8	CLK I/O	If CLK SEL is low then this pin inputs external clock source. If CLK SEL is high then this pin outputs internal clock pulses.			
9	CLK SEL	Clock select input. When low selects external clock source. When high selects internal clock source			
10	$\overline{\text{RD}}$	Active low read enable input if the display is selected, a low will enable the output drivers of the data bus.			
11	OA	OSC. ADJ. Oscillating Frequency can be adjusted by connecting this pin to VCC through an external resistor.			

CHARACTER SET

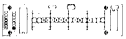
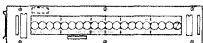
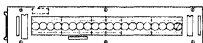
D0	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H
D1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
D2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
D3	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
D4	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
D5	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
D6	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
D7	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
HEX	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
L L H H L L	!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/	
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L H L L H H	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_

NOTES

- 1) A3 Must be held high to get into character set
- 2) All other inputs display Blank
- 3) When D7 is high, underline is enabled



LED Intelligent Display Assemblies

Pack- age Type	Package Outline	Part Number	Char- acter Height	Description	Page
16 Char. Assembly		IDA-1414-16	.112"	16 character assembly containing four DL-1414 displays	105
32 Char. Assembly		IDA-1416-32	.160"	32 character assembly containing eight DL-1416 displays	109
16 & 32 Char. Assembly		IDA-2416-16 IDA-2416-32	.160"	16 character assembly containing four DL-2416 displays 32 character assembly containing eight DL-2416 displays	113
16, 20 & 32 Char. Assembly		IDA-3416-16 IDA-3416-20 IDA-3416-32	.225"	16 character assembly containing four DL-3416 displays 20 character assembly containing five DL-3416 displays 32 character assembly containing eight DL-3416 displays	117
16 & 32 Char. Assembly		IDA-3422-16 IDA-3422-32	.170"/ .100"	16 character assembly containing four DL-3422 displays 32 character assembly containing eight DL-3422 displays. Phasing out—Not recommended for new designs.	121
16 & 20 Char. Assembly		IDA-7135-16 IDA-7137-16 IDA-7135-20 IDA-7137-20	.68"	16 character, 5x7 dot matrix assembly containing 16 DL-713X displays. Orange. 16 character, 5x7 dot matrix assembly containing 16 DL-713X displays. Green. 20 character, 5x7 dot matrix assembly containing 20 DL-713X displays. Orange. 20 character, 5x7 dot matrix assembly containing 20 DL-713X displays. Green.	125